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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MTH 205 — Past Questions & Materials

by Renance

Course: MTH 205

Institution: Federal University of Petroleum Resources, Effurun

Papers in this pack: 3

Levels: 200

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Paper 1 of 3 — MTH 205(222) PQ 2025

Level 200 • Second Semester • PastQuestion

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re; COLLEGE OF SCIENCE
ees, DEPARTMENT OF MATHEMATICS
SECOND SEMESTER EXAMINATION, 202€/202ACADEMIC SESSION
LEVEL: 200
COUSE: MTH222 (LINEAR ALGEBRA II) MTH Dos

Instruction: Answer any Four(4) questions

Time Allocated: 2hours

- Question |

A. (i) Define a linear equation. (3mks)

(ii) Differentiate between a linear function and a non-linear function, giving examples

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in each case. (3 r mks)

B. Explain the term “ Consistency “of a system of equations. (3mks)

C. (i) Test for the Consistency of the system of equations below;

$$5x, +4x, = 14$$

$$3x, + 2x, =8$$

$$-x, -X2 = -1 \text{ (4mks)}$$

“<-Kii) For what value of K is the system below consistent?

$$LEE \ 3x+y=-2$$

$$4x +2y=k$$

$$2x -3y = -3k \text{ (4mks)}$$

- Question 2

Using the Gaussian

below.

$$(a) \ 4x, + 3x2 + 2x3 - \%q = +$$

$$5x, + 4x2 + 3x3 - X4 =4$$

$$-2x, - 2X2 - X3 \text{ ody} = -3$$

$$11x, + 6x2 + 4x3 + \%4 = 11$$

Elimination Method, solve the Systems of equations in (a) and (b)

(7mks)

$$(b) \ x, + 2x3 + 4x4 = -8$$

$$Xz - 3X3 - X4 = 6 \text{ F}$$

$$3x, + 4x2 - 6X3 + 4\% = \text{ig aa}$$

Use any method of your choice to evaluate the system below.

$$Xz + 2x4 + 3X5 =0$$

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$$4x_1 + 8x_2 = 0 \text{ (Ga mks)}$$

$$\text{ot } iE) = P$$

xe ro

$$A \sim 1s \#$$

~- Question 3

$$A. \text{ Given the System: } x_1 + x_2 + 2x_3 = 8$$

$$-x_1 - 2x_2 + 3x_3 = 1$$

$$3x_1 - 7x_2 + 4x_3 = 10$$

(i) Evaluate "A7?", where A is the Augmented matrix of the system above. (7mks)

(ii) Find the Solution Set of the giver system. (45 mks)

B. Find the matrix representation of the linear operator "T " on "IR?" relative to the basis $f_1 = (1,0)$, and $f_2 = (0,1)$, given $T(x, y) = (6x, 3x - 4y)$. (6mks)

Question 4

A. Consider 2 regular $x'y'$ —Coordinate system, obtained by rotating a rectangular xv - Coordinate system through an angle; $d = >$ Then evaluate: —

(i) The $x'y'$ — Coordinate of the point whose xy - Coordinates are $(-1, 3)$. (4mks)(ii) The xy — Coordinate of the point whose $x'y'$ -Coordinates are $(3, 2)$. (4 mks)

B. Prove that the Similarity of Matrices is an Equivalence Relation. (9mks)

, Question 5

2 L 7

suppose $A = \begin{pmatrix} 1 & 2 \\ i & \end{pmatrix}$

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(a) Verify the Caley Hamilton Theorem. (5mks)

(b) Evaluate the Characteristic Roots of A. (25 mks)

(c) Evaluate the Characteristic Vectors of A. (5mks)

(c) Find the Minimum Polynomial of A. (5mks)

Question 6

A. Explain the following terms +

(i) Homogeneous Quadratic Form (3mks) =

(ii) Symmetric Bilinear Form (3mks) :

(iii) Quadratic Form (3mks) las

(iv) Canonical Form (3mks)

B. Suppose T is a Bilinear Form on $\mathbb{R}^2$, defined by $T(x,y) = 6x_1^2 + 4x_1x_2 + 2x_2^2 + 3x_2y_2$, where $x = (x_1, x_2)$, and $y = (y_1, y_2)$, Find the matrix of T in the Basis $e_1 = (1,2)$ and $e_2 = (1,-2)$. (5+mks) |

Paper 2 of 3 — MTH 205 or 225 Real Analysis Second Semester 2020 2021

Level 200 • Second Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF SCIENCE
DEPARTMENT OF MATHEMATICS

SECOND SEMESTER EXAMINATION 2020/2021

Course Code/Title: MTH 225-Real Analysis II

Instructions: Attempt any 4 questions. Time Allowed: 2½ Hours.

Question 1 (17 ½ Marks)

- (a) Define clearly Pointwise convergence of sequence of rvfs (3 marks)
- (b) For each integer $n \geq 1$, define $f_n(x) = ne^{-nx^2}$ on $[0, 1]$, prove or disprove that $\{f_n\}_{n \geq 1}$ converges pointwise on $[0, 1]$ (8½ marks)
- (c) Sketch the first 3 terms of $f_n(x) = ne^{-nx^2}$ (6marks)

Question 2 (17 ½ Marks)

- (a) State the Uniform Cauchy Criterion theorem (4marks)
- (b) Let $f_n(x) = \frac{x^n}{5+x} \forall x \in [0, 4], n \geq 1$. Find the set D a subset of $[0, 4]$ that well defines the sequence of real valued function. (5marks)
- (c) Show that $f_n(x) = \frac{x^n}{5+x} \forall x \in [0, 1]$ is uniform convergence on $[0, 1]$. (8½ marks)

Question 3 (17 ½ Marks)

- (a) State the alternative test of uniform convergence of sequence of functions (3marks)
- (b) Show that $f_n: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f_n(x) = \frac{2n+\sin nx}{3n+\sin^2 nx}$, for $x \in \mathbb{R}$ converges on $\mathbb{R}$. (8 ½ marks)
- (c) If $f_n(x) = \frac{\sin nx}{\sqrt{n}}$ for each $n \in \mathbb{N}$ and $x \in [0, 2\pi]$, then find $\frac{d f_n(x)}{dx}$ and show that it does not converge to $\frac{d f(x)}{dx} \in [0, 2\pi]$. (6 marks)

Question 4 (17 ½ Marks)

- a) When is a bounded function $f: [a, b] \rightarrow \mathbb{R}$ said to be Riemann Integrable? (4marks)
- b) Prove that the function $f: [0, 1] \rightarrow \mathbb{R}$ defined by $f(x) = x^3$ for all $x \in [0, 1]$ is Riemann Integrable and that $\int_0^1 f(x) dx = \frac{1}{4}$ (10 ½ marks)
- c) State three (3) basics properties of Riemann Integral (3marks)

Question 5 (17 ½ Marks)

- (a) State and prove Weierstrass M. Test (10marks)
- (b) What is (i) Harmonic series (ii) Alternating Series (iii) P-series (5 ½ marks)
- (c) Give examples of each type in 5(b) (2 marks)

Question 6 (17 ½ Marks)

- a) Define the term Power series and give 2 examples (5marks)
- b) Define radius of convergence. Use your definition to find the radius of convergence of the one of the example of power series you gave in 6(a) (9marks)
- c) State without prove Weierstrass M. Test (3 ½ marks)

Paper 3 of 3 — MTH 205 Vector Analysis First Semester 2018/2019

Level 200 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN COLLEGE.

DEPARTMENT OF MATHEMATICS & COMPUTER SCIENCE

FIRST SEMESTER EXAMINATION 2018/2014 SESSION

COURSE CODE:MTH 215 ;COURSE TITLE: VECTOR ANALYSIS

INSTRUCTIONS: ANSWER QUESTION 1 AND ANY OTHER TWO: TIME 2 HOURS

ill. Show that vector $(\sin \theta \sin \phi, \sin \theta \cos \phi, \cos \theta)$ has unit magnitude for any θ and ϕ .

$\cos \theta$

(b) Let $b(t)$ be a unit vector. Show that $\dot{b}$ is always perpendicular to b if $b \neq 0$

(c) i. Let $p(x, y, z)$ be a continuously differentiable scalar field. Define its gradient.

ii. Find ∇p , where $p(x, y, z) = x^2 + y^2 + z^2$

(d) 1. Let $\phi(x, y)$ be a continuously differentiable function of x and y . Consider a point $P(a, b)$ on the (x, y) plane, introduce a unit vector u making angle θ with the axis. Define the directional derivative of $\phi(x, y)$ at point $P(a, b)$ in direction u , $D_u \phi(a, b)$

ii. Show that $D_u \phi(a, b) = \nabla \phi(a, b) \cdot u$

QUESTION TWO (20 marks)

(a) The position vectors of points P and Q are given by $r_P = 2i + 3j - k$, $r_Q = 4i - 3j + 2k$. Determine PQ in terms of i, j, k and find its magnitude.

(b) If $A = 3i - j - 4k$, $B = -2i + 4j - 3k$ and $C = i + 2j - k$, find: $1. 2A - B + 3C$

$|A - B + C|$ $2. A$ unit vector parallel to $3A - 2B + 4C$

QUESTION THREE (20 marks)

(a) 1. Prove that $A \cdot B = B \cdot A$

ii. Prove that $B \times A = -A \times B$

(b) If $A = A_1i + A_2j + A_3k$ and $B = B_1i + B_2j + B_3k$.

Prove that $A \cdot B = A_1B_1 + A_2B_2 + A_3B_3$

QUESTION FOUR (20 marks)

(a) Find the directional derivative of $\phi = 4xz^2 - 3x^2y^2z$ at $(2, -1, 2)$ in the direction $2i - 3j + 6k$.

(b) Find the directional derivative of $P = 4e^{xyz} + x^2$ at the point $(1, 1, -1)$ in a direction towards the point $(-3, 5, 6)$.

QUESTION FIVE (20 marks)

Given that $R = \sin t i + \cos t j + tk$, find:

$\frac{dR}{dt}$, $\frac{d^2R}{dt^2}$

(a) $\frac{dR}{dt} \cdot \frac{dR}{dt}$